

We can solve this homogeneous ODE with integration

$$\int_{T_{in}^*}^{T_{out}^*} \frac{dT^*}{T^*} = - \frac{hPL}{\rho A C \bar{u}} \int_0^L dx^*$$

$$\ln \left(\frac{T^*}{T_{in}^*} \right) + \frac{hPLx^*}{\rho A C \bar{u}} = 0$$

$$T^* = T_{in}^* e^{-\frac{hPLx^*}{\rho A C \bar{u}}} \Rightarrow T_{in}^* = \frac{T_{in} - T_w}{T_{in} - T_w} = 1$$

$$T_f = T_w + (T_{in} - T_w) e^{-\frac{hPx}{\rho A \bar{u} C}}$$

↳ Bulk fluid temperature as a function of x along the length of the rectangular channel.

Also remember: $T_f = T_b =$ bulk fluid temperature

$$\text{i.e. } T_b = \frac{1}{\rho A \bar{u} C} \int_A \rho C u(z, y) \cdot T(z, y) dA$$

$$\text{Note also: } dQ = hP dx (T_w - T_b) \quad \downarrow \text{ or } T_f$$

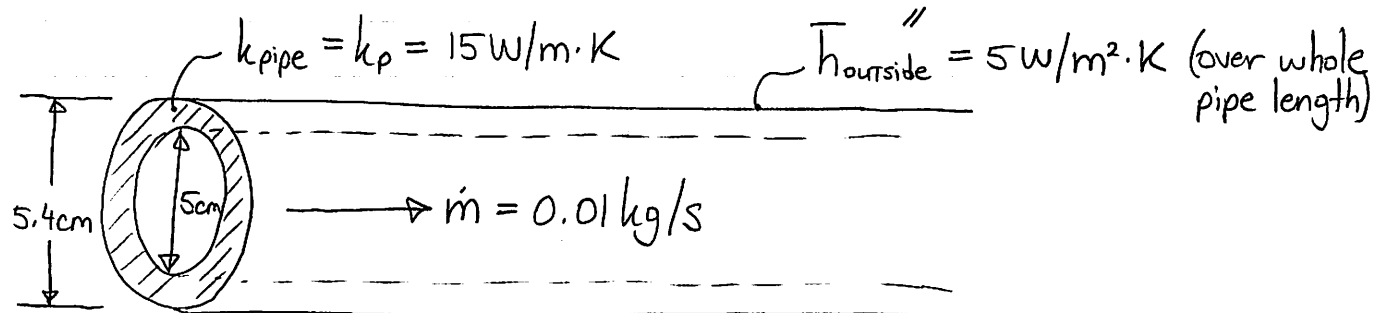
↳ $h = f(k_f) = f(T_b) \Rightarrow$ fluid properties are temperature dependent

For problems like these (internal flow), evaluate properties at $\bar{T}$

$$\bar{T} = \frac{T_{in} + T_{out}}{2}$$

Example | Let's do a comprehensive example that uses everything we've learned so far in class (conduction & convection).

Freezing water pipes in New England: $\bar{h}_o$



$$\sim T_{\infty} = -10^{\circ}\text{C}$$

$T_{\text{in}} = 1.5^{\circ}\text{C}$ (inlet pipe fluid bulk temperature)

- (a) Calculate the heat transfer coefficient on the inside of the pipe ($\bar{h}_i$). Assume fully developed flow (both thermally and hydrodynamically), and no ice is present.

First let's see our flow regime:

$$\dot{m} = \frac{\rho_w \pi D_{\text{in}}^2 \bar{U}}{4} \Rightarrow \text{from tables, } \rho_w = 1000 \text{ kg/m}^3$$

$$U_w = 1.5 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k_w = 0.6 \text{ W/m}\cdot\text{K}$$

$$Re_o = \frac{\bar{U} D_{\text{in}}}{U_w} = \frac{4 \dot{m}}{\pi \rho_w D_{\text{in}} U_w} = 170 \text{ (LAMINAR FLOW)}$$

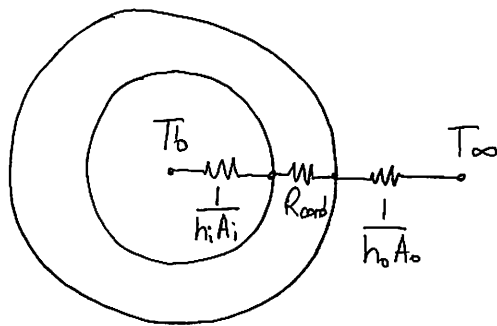
So we know $\overline{Nu}_o = \text{constant}$, however we don't have constant wall temperature or constant heat flux.

As a rough approximation, we can use the average:

$$Nu_o \approx \frac{1}{2} (3.657 + 4.364) = 4.01 \Rightarrow \boxed{\bar{h}_i = \frac{\overline{Nu}_o \cdot k_w}{D_{\text{in}}} = 45.7 \text{ W/m}^2\cdot\text{K}}$$

- (b) Using our calculated $\bar{T}_h$, determine the bulk water temp. T_b at the point at which water first freezes.

To solve this question, we need to understand the overall heat transfer from the outside air, to the bulk fluid.



Overall thermal resistance

$$\frac{1}{\pi D_{in} U} = \frac{1}{\pi D_{in} \bar{h}_i} + \frac{\ln(D_{out}/D_{in})}{2\pi k_p} + \frac{1}{\pi D_{out} \bar{h}_o}$$

We can solve for the overall heat transfer coefficient U

$$U = 4.20 \text{ W/m}^2 \cdot \text{K}$$

Now we can solve for when water begins to freeze:

$$q = \bar{h}_i (T_b - T_{freeze}) = U (T_b - T_{\infty})$$

$\uparrow$
 $T_{freeze} = 0^\circ \text{C}$

Rearranging:

$$T_b = \frac{\bar{h}_i T_{freeze} - U T_{\infty}}{\bar{h}_i - U} = 1.20^\circ \text{C}$$

(c) At what point in the pipe length L does the water begin to freeze?

Applying an energy balance on a differential fluid element just like the previous example:

$$\dot{m}c \frac{dT_b}{dx} = \pi D_{in} U (T_{\infty} - T_b)$$

We already solved that:

$$\frac{T_b - T_{\infty}}{T_{in} - T_{\infty}} = \exp\left(-\frac{\pi D_{in} U L}{\dot{m}c}\right); \text{ note } T_{in} = \text{inlet fluid temperature}$$

From this we can solve for L given that $T_b = 1.20^{\circ}\text{C}$
 $T_{\infty} = -10^{\circ}\text{C}$

$$L = \frac{\dot{m}c}{\pi D_{in} U} \cdot \ln\left(\frac{T_{in} - T_{\infty}}{T_b - T_{\infty}}\right) = 1.47 \text{ m}$$

$T_{in} = 1.5^{\circ}\text{C}$
 $C = 4200 \text{ J/kg} \cdot \text{K}$
(from Tables)

(d) Suppose the flow rate is increased to $\dot{m} = 1 \text{ kg/s}$. At a point $L = 10 \text{ m}$ from the entrance, determine the temperature on the inner wall of the pipe. Assume same conditions as before.

First let's check our flow regime again:

$$Re_{Dn} = \frac{4\dot{m}}{\pi \rho_w D_{in} U_w} = 1.7 \times 10^4 > 2300 \text{ (Turbulent)}$$

So we need to use the Gnielinski correlation:

$$f = (0.79 \ln Re_0 - 1.64)^{-2} = 0.0273$$

$$\overline{Nu}_0 = \frac{h_i D_{in}}{k_f} = \frac{(f/8)(Re_0 - 1000) Pr_w}{1 + 12.7 (f/8)^{1/2} (Pr_w^{2/3} - 1)}$$

From Tables, $Pr_w \cong 11$

$$\overline{Nu}_0 = 153 = \frac{\overline{h}_i D_{in}}{k_f} \Rightarrow \overline{h}_i = 1610 \text{ W/m}^2 \cdot \text{K}$$

We can now re-calculate our overall heat transfer coefficient, U :

$$U = \left\{ \pi D_{in} \left[\frac{1}{\pi D_{in} \overline{h}_i} + \frac{\ln(D_{out}/D_{in})}{2\pi k_p} + \frac{1}{\pi D_{out} \overline{h}_o} \right] \right\}^{-1}$$

$$U = 5.38 \text{ W/m}^2 \cdot \text{K}$$

Now we use our energy balance equation to solve for T_b at $L = 10 \text{ m}$

$$T_b = T_\infty + (T_{in} - T_\infty) \exp\left[-\frac{\pi D_{in} U L}{\dot{m} c}\right] = 1.48^\circ \text{C}$$

Finally, using thermal resistance to solve for inner surface temperature: $(T_{s,i})$

$$q = \overline{h}_i (T_b - T_{s,i}) = U (T_b - T_\infty)$$

$$T_{s,i} = \frac{(\overline{h}_i - U) T_b - U T_\infty}{\overline{h}_i} = 1.44^\circ \text{C}$$

So we see that one mitigation strategy to prevent icing is to increase the flow rate.